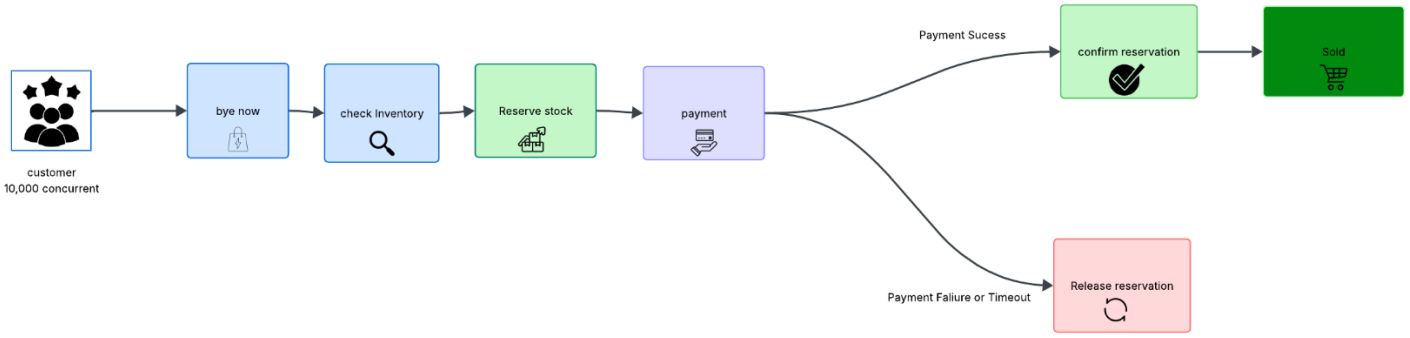
SALESTORM

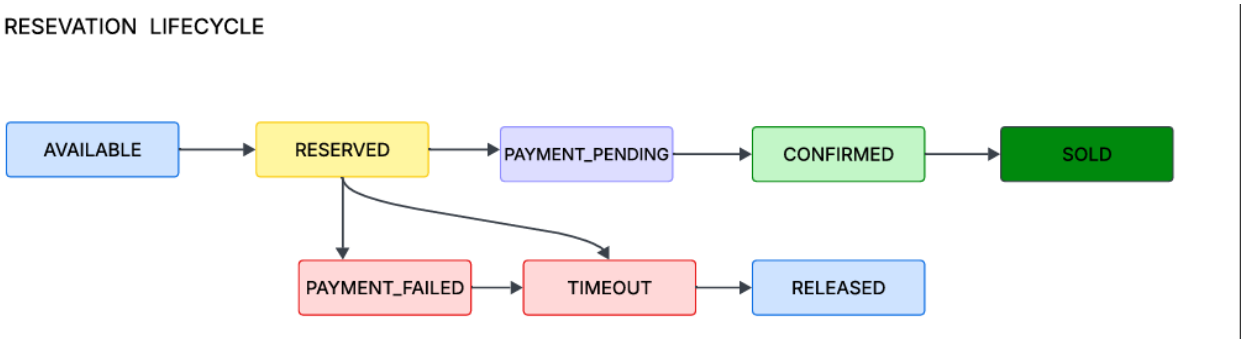
High-Scale E-Commerce Scale

Stage 3 – Concurrency & Inventory Design

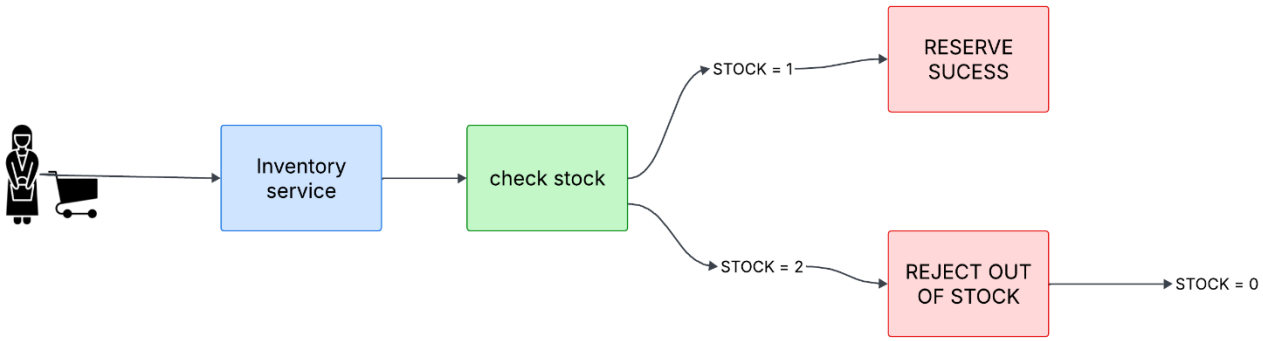
1) Design a concurrency-safe inventory reservation mechanism.



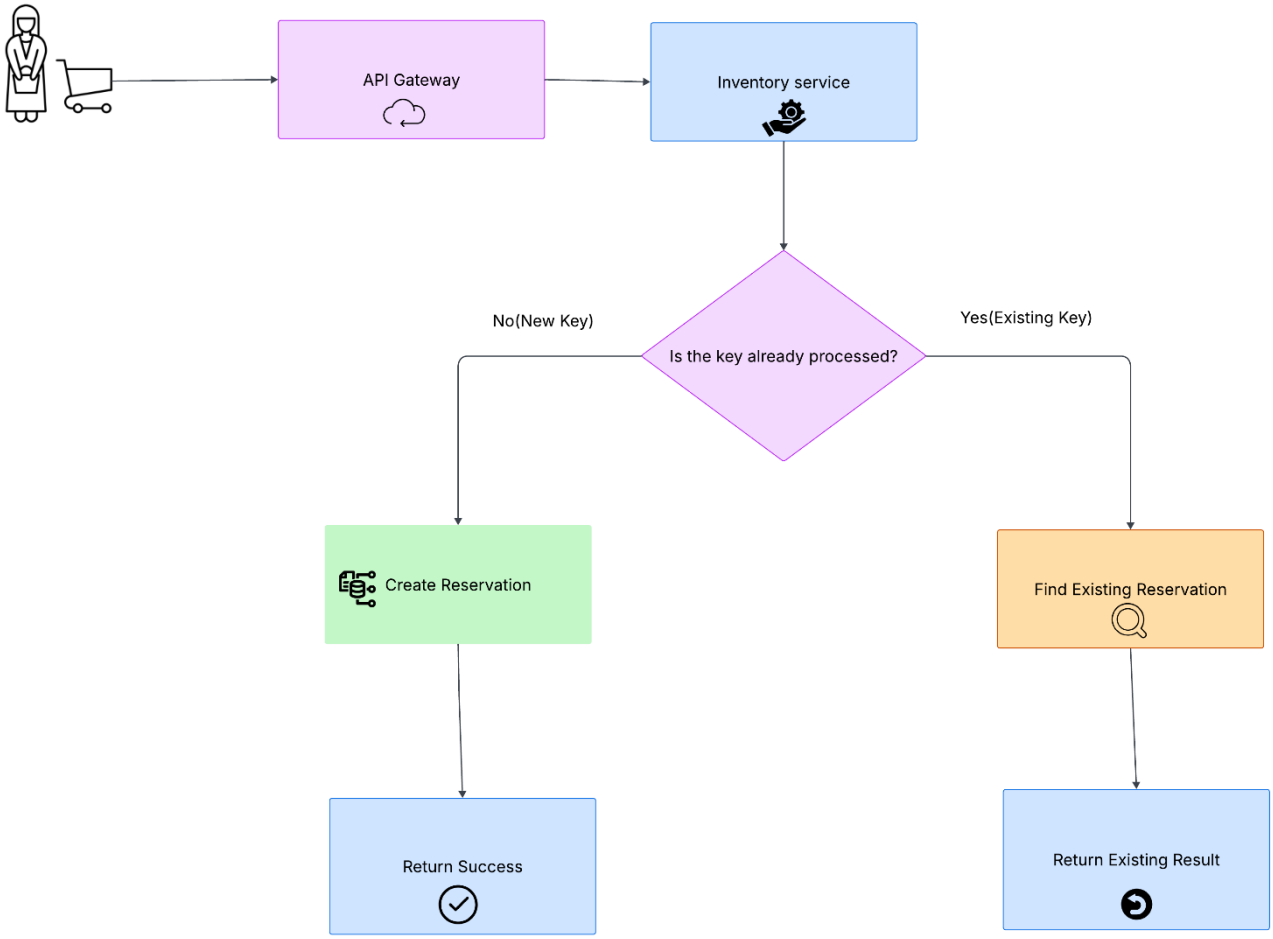
2) Define reservation creation, confirmation, expiry and release.



3) Prevent overselling when multiple customers request the last available units.



4) Handle duplicate reservation requests.



5) Compare at least two concurrency-control approaches and justify the selected approach.

Approach	**Key Points**
Optimistic	High throughput, no locks, but more conflicts and retries under heavy traffic
Pessimistic	Locks inventory during update, prevents conflicting reservations, but requests may wait

6) Define transaction boundaries and consistency guarantees

TRANSACTION BOUNDARIES

Reserve Inventory (Atomic Transaction)

1 . Check available quantity

2. Valid request

3.Reduce Available Quantity

4.Increase reserved quantity

5.Create reservation record

Payment & Confiramation

1. Payment happens

2.confirm

3.Update inventory

Update order Status